

TEAM-DEVELOPED ROCKET AND FLIGHT COMPUTER

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Tech Stack: Embedded C++, ESP32-S3, SOLIDWORKS, OpenRocket

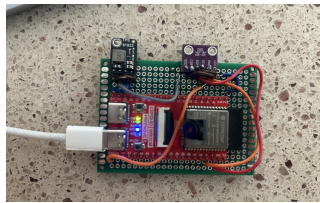
OBJECTIVE: Engineered and launched a custom high-power rocket with a self-built ESP32-S3 flight computer, 3D-printed aerostructures, and a custom ignition and safety-lock system.



Final Assembled Rocket

DEVELOPMENT WORKFLOW

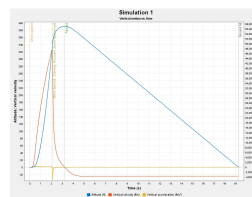
- **1. Flight Computer:** Engineered and coded a custom ESP32-S3 flight computer with real-time SD data logging.
- **2. Aerodynamics:** Modeled and validated flight stability using OpenRocket simulations.
- **3. Sensor Integration:** Integrated a BME280 pressure sensor and TPS61023 boost converter for altitude tracking and stable power delivery.
- **4. 3D-Printed Parts:** Designed and fabricated custom fins and nose cone in SOLIDWORKS.
- **5. Ignition & Recovery:** Engineered a custom ignition circuit and parachute recovery system from the ground up.
- **6. Flight Testing:** Ran iterative test launches to validate flight computer and recovery performance.



ESP32-S3 Flight Computer



3D-Printed Fin & Motor Mount



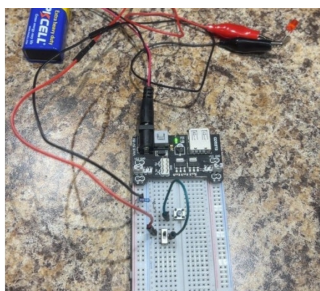
OpenRocket Flight Simulation



Rocket Setup

ENGINEERING CHALLENGES & SOLUTIONS

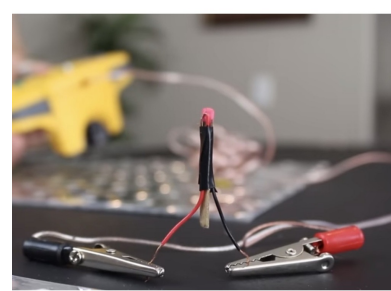
- **Custom Ignition System:** Built and tested our own motor igniter circuit rather than using a pre-made one.
- **Launch Safety Interlock:** Engineered a key-locked launch interlock, adding a dedicated safety stage to prevent accidental ignition.
- **Recovery System:** Designed and validated parachute deployment and canopy geometry for a safe, controlled descent.
- **Full Integration:** Unified custom firmware, sensors, and recovery hardware into one fully integrated, flight-ready system.



Safety Lock Prototype



Recovery Parachute



Custom-Built Igniter